

Biological Dimensions and Their Role in Achieving Food Security and Economic Development

الأبعاد البيولوجية ودورها في تحقيق الأمن الغذائي والتنمية الاقتصادية

Khadidja Madani¹, Amina Zoubida Madani²

¹ University Ain Temouchent Belhadj Bouchaib (Algeria), dr_madani@hotmail.fr

² Saad Dahlab University, Blida 1 (Algeria), amina-hydro@hotmail.com

Received: 06/09/2025

Accepted: 28/12/2025

Published: 05/01/2026

Abstract:

Livestock in Algeria is a cornerstone for achieving food security and a key driver of economic development, contributing significantly to the agricultural GDP and providing essential animal protein for consumers. Data from 2014–2024 show a notable improvement in economic indicators, with the total value of livestock production and its share in agricultural GDP steadily increasing, supported by government programs aimed at enhancing productivity and improving breeds.

From a biological perspective, the numbers of cattle, sheep, and goats, along with milk, meat, and egg production, have shown gradual growth despite climatic challenges and occasional feed shortages. This reflects the sector's resilience and capacity for sustainable growth, reinforcing its strategic role in supporting the national economy and meeting the growing food demand.

Keywords: Livestock; Food security; Economic development; Animal production; Biological indicator; Local breeds; Algeria.

JELClassificationCodes: Q18, Q12, O13, Q57.

ملخص:

تعد الثروة الحيوانية في الجزائر ركيزة أساسية لتحقيق الأمن الغذائي ودعم مهمة للتنمية الاقتصادية، حيث تسهم بشكل كبير في الناتج المحلي الزراعي وتوفير البروتين الحيواني للمستهلك. أظهرت البيانات للفترة 2014–2024 تطوراً ملحوظاً في المؤشرات الاقتصادية، إذ ارتفعت القيمة الإجمالية للإنتاج الحيواني ومساهمته في الناتج المحلي الزراعي بشكل تدريجي، مدعومة ببرامج حكومية لتحسين الإنتاجية وتطوير السلالات. وعلى الصعيد البيولوجي، شهدت أعداد الأبقار والأغنام والماعز، إضافة إلى إنتاج الحليب واللحوم والبيض، نمواً تدريجياً رغم التحديات المناخية ونقص الأعلاف في بعض السنوات، وهو ما يعكس قدرة القطاع على التكيف والنمو المستدام، مما يعزز دوره الاستراتيجي في دعم الاقتصاد الوطني وتلبية الطلب الغذائي المتزايد.

كلمات مفتاحية: الثروة الحيوانية؛ الأمن الغذائي؛ التنمية الاقتصادية؛ الإنتاج الحيواني؛ السلالات المحلية؛ المؤشرات البيولوجية؛ الجزائر.

تصنيفات JEL : Q18، Q12، O13، Q57.

¹ Corresponding author: Khadidja Madani, e-mail: dr_madani@hotmail.fr

1. INTRODUCTION

Livestock in Algeria represents a cornerstone of the agricultural system, contributing directly to food security by providing essential products such as meat, dairy, and leather. Moreover, it plays a vital role in enhancing GDP and supporting economic activity, particularly in rural areas. Algeria possesses natural and human resources that enable the development of this sector, including vast grazing lands, environmental diversity, and a growing domestic demand for animal products. However, the sector faces biological, technical, and economic challenges related to productivity, feed resources, and animal diseases. Addressing these issues requires comprehensive strategies that integrate improvements in biological aspects with infrastructure development and better marketing mechanisms.

In recent years, particularly during the period 2014–2024, statistical data have shown significant fluctuations in production levels and economic contributions of the livestock sector. While milk and meat production experienced periods of decline due to feed shortages and climatic pressures, government programs aimed at genetic improvement, veterinary services, and production support have contributed to gradual recovery and resilience. The increase in small ruminant production and egg output during this decade highlights the biological adaptability of local breeds and the potential of the sector to respond to rising food demand.

Thus, the period 2014–2024 provides a clear picture of both the vulnerabilities and strengths of livestock development in Algeria. It reflects not only the sector's central role in securing the country's nutritional needs but also its capacity to act as a driver of sustainable economic development when supported by appropriate policies and innovations.

1.1 Research Problem

Despite Algeria's large livestock population, national production fails to meet domestic food needs. What biological and economic barriers prevent this sector from becoming a strong contributor to the national economy?

1.2 Research Objectives

- ✓ Assess the biological status of livestock, including health and genetic potential.

- ✓ Identify obstacles related to production, nutrition, and infrastructure.
- ✓ Propose policies to enhance national productivity and reduce reliance on imports.

1.3 Research Significance

- ✓ Highlight a vital food resource and key economic pillar, while offering applicable recommendations to move the livestock sector toward self-sufficiency.
- ✓ Emphasize the biological dimensions of livestock and their impact on improving the quality and quantity of food production.
- ✓ Analyze the current state of livestock in Algeria in terms of production, distribution, and challenges.
- ✓ Evaluate the sector's actual contribution to food security and narrowing the food gap.
- ✓ Provide a future-oriented vision based on innovation and modern technologies in breeding and production.
- ✓ Formulate practical recommendations to achieve self-sufficiency and strengthen the livestock sector's role in the national economy.

1.4 Research Structure

Section One: The Biological Dimensions of Livestock in Algeria

Section Two: The Role of Livestock in Food Security and Economic Development.

2. The Biological Dimensions of Livestock in Algeria

The biological dimensions of livestock in Algeria are reflected in the wide diversity of species and their distribution across different regions. The country hosts millions of heads of sheep, goats, cattle, camels, and horses, with unique local breeds such as the Ouled Djellal sheep, renowned for its adaptability to harsh climatic conditions. Livestock also plays an important role in maintaining ecological balance; however, it faces growing threats such as climate change, overgrazing, poaching, and the loss of natural habitats. To mitigate these risks, the state has adopted national programs that include precise herd census, genetic improvement of breeds, feed subsidies, and raising environmental awareness among breeders, thereby ensuring the long-term sustainability of these vital resources.

In addition, the livestock sector contributes significantly to the preservation of biodiversity by maintaining indigenous breeds that are uniquely adapted to local ecosystems. These breeds not only ensure genetic richness but also represent a valuable resource for future crossbreeding and productivity improvement programs. Recent studies indicate that Algeria's diversified geography, from the humid north to the arid Sahara, provides a natural laboratory for observing the interaction between animals and their environment, reflecting resilience strategies in the face of climatic pressures.

Furthermore, the biological aspect of livestock is closely connected with socio-economic realities. For many rural households, small ruminants such as sheep and goats are not merely a source of food, but also a form of financial security and cultural heritage. The sustainability of these resources requires integrated policies that combine scientific research, modern breeding techniques, and traditional knowledge, ensuring that biological diversity is preserved while meeting the growing demands of food security and economic **(Brown, 2006, p. 35)**

2.1 Animal Diversity, Breeding, and Veterinary Services

✓ Animal Diversity and Geographical Distribution

Algeria hosts a rich variety of livestock species: local cattle breeds (such as Guelma and Chaouia), imported high-yield breeds (Holstein, Montbéliarde), sheep breeds such as Ghannami and Hamraoui, goats resistant to desert conditions, camels in the southern regions, and poultry concentrated in urban areas.

Cattle are mainly found in the humid northern regions, while sheep and goats are concentrated in the high plateaus and the south, reflecting the ecological adaptation of each species.

The geographical distribution of these animals is strongly shaped by Algeria's ecological diversity. Cattle are predominantly raised in the humid and sub-humid northern regions, where pastures and water resources are abundant. Sheep and goats, however, are concentrated in the high plateaus and steppe zones, where their hardiness allows them to survive in harsh climates with limited feed resources. Camels are exclusive to the Saharan areas, reflecting their ecological specialization in extreme desert environments. Poultry farms, benefiting from proximity to markets, are

clustered around large cities such as Algiers, Oran, and Constantine.

This spatial diversification not only reflects ecological adaptation but also enhances resilience to climate variability by spreading production across different ecosystems. For instance, if drought affects cereal production in the north, sheep and goat herding in the south and steppe areas can still provide a stable source of protein. Moreover, the integration of livestock with crop farming systems contributes to sustainability through manure recycling, crop residue utilization, and the preservation of grazing lands.

In short, the heterogeneity of Algeria's livestock resources provides a strong foundation for both ecological balance and economic growth. By capitalizing on this diversity through genetic improvement programs, better feed management, and stronger veterinary services, Algeria can significantly increase the efficiency and sustainability of its livestock sector while reinforcing its contribution to food security and rural development. **(Foreign Agricultural Service, 2022).**

✓ **Genetic Improvement and Breeding**

Algeria has launched artificial insemination programs and the importation of pregnant heifers to enhance productivity.

For example, the Holstein breed can produce between 5,000–7,000 liters of milk per year, compared to only 1,200–1,800 liters for local breeds. However, actual production remains strongly dependent on feed availability and infrastructure conditions. **(Dairy Global, 2019)**

In addition to Holstein, other high-yield breeds such as Montbéliarde have been introduced to improve both milk and meat productivity. Selective breeding programs aim to combine the adaptability of local breeds with the productivity of imported ones, thereby creating crossbreeds that are more resistant to diseases and environmental stress while still achieving higher yields.

The government has also invested in the establishment of genetic banks and semen storage facilities to preserve biodiversity and maintain a reliable supply of genetic material for future breeding initiatives. Moreover, training centers have been set up to educate farmers on modern breeding techniques, reproductive health management, and the importance of recording animal lineage for traceability.

Beyond dairy production, genetic improvement initiatives target sheep and goat breeds, focusing on meat quality, wool production, and resistance to arid climates. In southern Algeria, for instance, camel breeding programs are being developed to enhance milk yield and adapt to international demand for camel-based products.

Looking ahead, the integration of biotechnology and genomic selection is expected to revolutionize the breeding sector in Algeria, allowing for more precise improvements in productivity, disease resistance, and climate adaptability. Such advancements not only improve food security but also enhance the competitiveness of Algerian livestock in regional and global market

✓ **Animal Health and Veterinary Services**

The country implements national vaccination programs against diseases such as foot-and-mouth disease.

Additionally, a digital identification system and mobile veterinary centers have been introduced to improve livestock tracking and provide veterinary support in remote areas.

Moreover, the integration of information and communication technologies in veterinary services has enabled better monitoring of animal movements, ensuring biosecurity and facilitating rapid intervention in case of disease outbreaks. The use of mobile veterinary units has proven particularly effective in reaching pastoral communities in the High Plateaus and the Sahara, where access to fixed infrastructure is limited.

At the same time, training programs for breeders and veterinary staff have been strengthened to raise awareness about disease prevention, animal welfare, and hygiene standards. Such initiatives not only improve productivity and food safety but also enhance the competitiveness of Algerian livestock products in both domestic and international markets. In the long term, these efforts are expected to build a more

Furthermore, the promotion of public–private partnerships in veterinary services has created new opportunities for expanding coverage and ensuring sustainable funding mechanisms. For example, private veterinary clinics are increasingly complementing public services by offering specialized care, artificial insemination, and reproductive health programs that directly

improve herd productivity.

In recent years, Algeria has also intensified its efforts to combat transboundary animal diseases by establishing quarantine zones at borders, enforcing strict sanitary regulations for animal trade, and cooperating with neighboring countries to harmonize disease control strategies. This regional cooperation has become essential in preventing the spread of epidemics such as peste des petits ruminants (PPR) and avian influenza, which threaten both food security and rural livelihoods.

Additionally, new investments have been directed toward the modernization of veterinary infrastructure, including the renovation of national veterinary schools, the establishment of specialized research centers, and the development of digital databases for disease surveillance. These initiatives aim to improve the quality of veterinary education and create a new generation of professionals capable of addressing the complex challenges facing the livestock sector.

Finally, the incorporation of the One Health approach is gaining momentum, recognizing the interconnection between animal health, human health, and environmental sustainability. By strengthening biosecurity, reducing the misuse of antibiotics, and promoting eco-friendly livestock practices, Algeria is not only protecting its herds but also contributing to global efforts in antimicrobial resistance control and sustainable development. (World Bank, 2023)

2.2 Feed Resources and Nutritional Challenges

✓ Feed Resources and Nutrition

- Algeria possesses vast grazing areas, yet they remain vulnerable to drought and desertification.
- There is increasing reliance on concentrated feed such as yellow corn and soybean meal.
- Ongoing research is exploring local alternatives such as hydroponic green fodder and irrigated barley (FAO, 2005)

3. The Role of Livestock in Food Security and Economic Development

Livestock represents a fundamental pillar for achieving food security and fostering sustainable economic development in Algeria. It provides

essential, high-nutritional-value products such as meat, milk, and their derivatives, while also serving as a supplier of raw materials for agro-industries. Beyond its nutritional contribution, the sector plays a central role in diversifying national income sources, generating employment opportunities, and stimulating both rural and urban economies.

Given these multidimensional benefits, the development of the livestock sector emerges as a strategic choice for ensuring food stability, supporting self-sufficiency, and enhancing the competitiveness of the national economy in the face of global challenges.

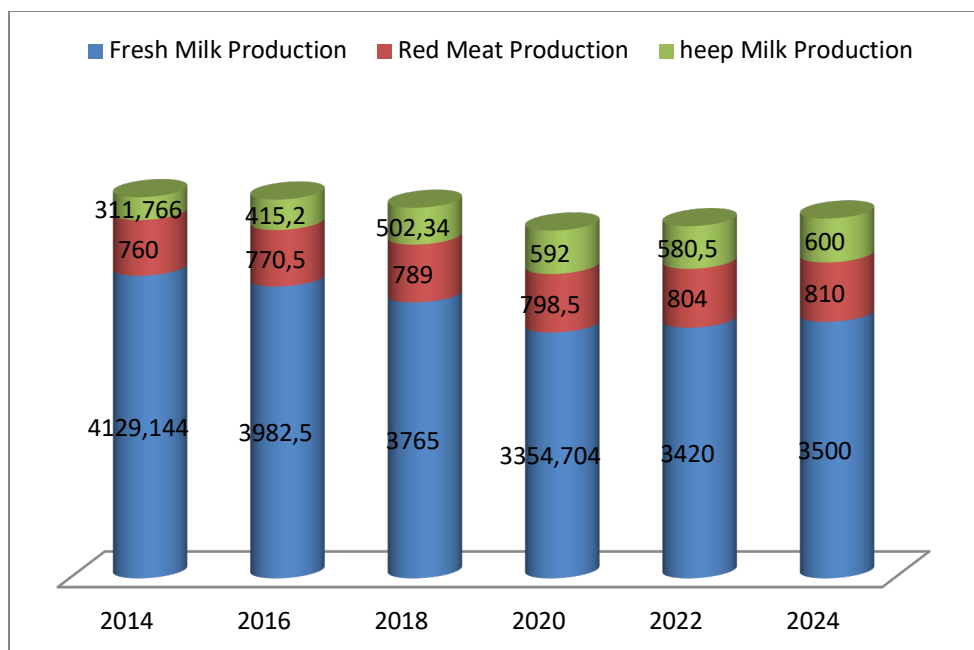
Ongoing research is increasingly focused on local alternatives to strengthen the sector's resilience, such as hydroponic green fodder and irrigated barley, which can mitigate feed shortages and reduce reliance on imported concentrates.

3.1 Food Security: Key Indicators (2014–2024)

Food security in Algeria is closely linked to the performance of the livestock sector, which provides a large share of the population's protein needs through meat, milk, and their derivatives. Monitoring key indicators such as production volumes, availability, and consumption trends is essential to assess the sector's ability to meet the growing food demand. The period 2014–2024 witnessed significant fluctuations in livestock production, reflecting both the sector's potential and the challenges it faces.

While supportive government programs aimed at increasing self-sufficiency have been implemented, factors such as climatic variability, feed shortages, and structural constraints have continued to affect output levels. Analyzing these indicators allows for a better understanding of the progress made toward food security and highlights the remaining gaps that need to be addressed to ensure long-term stability.

Fig.1. Food Security (in tons)



Source: Foreign Agricultural Service, 2022, p15.

Figure (01) illustrates the concept of food security in the Milk production in Algeria declined from over 4.1 million tons in 2014 to approximately 2.5 million tons in 2022, indicating a persistent food gap despite supportive government programs.

In terms of meat production, total output remained relatively stable at around 800,000 tons, with beef accounting for nearly 20% of the total.

Sheep milk production, however, showed remarkable growth, increasing from 311,000 tons in 2014 to more than 592,000 tons in 2020, reflecting the potential of small ruminants in bridging part of the food deficit. (Foreign Agricultural Service, 2022)

-Economic Development

-Livestock products (meat, milk, hides, wool) provide significant industrial and agricultural inputs for the country.

-Value chains in the sector generate employment opportunities for young people in breeding, processing, and rural distribution.

-Integration between agriculture and livestock is evident: animal manure is used to fertilize crops, while agricultural by-products serve as feed resources. **(Husfarm, n.d.)**

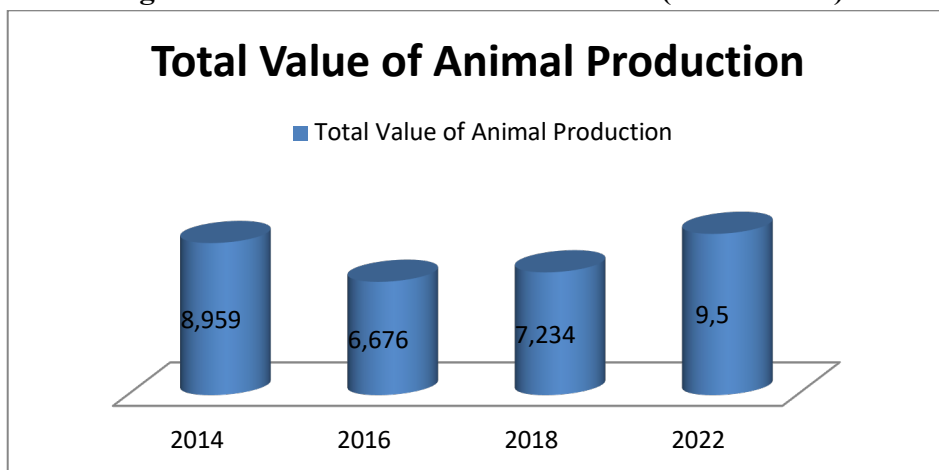
The livestock sector also contributes significantly to foreign exchange savings by reducing dependency on imported meat and dairy products. At the same time, it opens opportunities for export of niche products such as camel milk, organic honey, and high-quality sheep wool to international markets.

Rural development is strongly tied to livestock activities, as they ensure income diversification for households, improve resilience against economic shocks, and strengthen social stability in vulnerable regions.

Moreover, government support programs, such as subsidies for feed, credit facilities for breeders, and investments in infrastructure (slaughterhouses, cold storage, and transportation networks), enhance productivity and increase the competitiveness of Algerian livestock in both domestic and foreign markets.

In the long term, the livestock industry has the potential to become a driver of inclusive economic growth, fostering innovation through biotechnology, genetic improvement, and sustainable breeding practices, while also contributing to the achievement of the Sustainable Development Goals (SDGs).

Fig.2. Total Value of Animal Production (Billion USD)



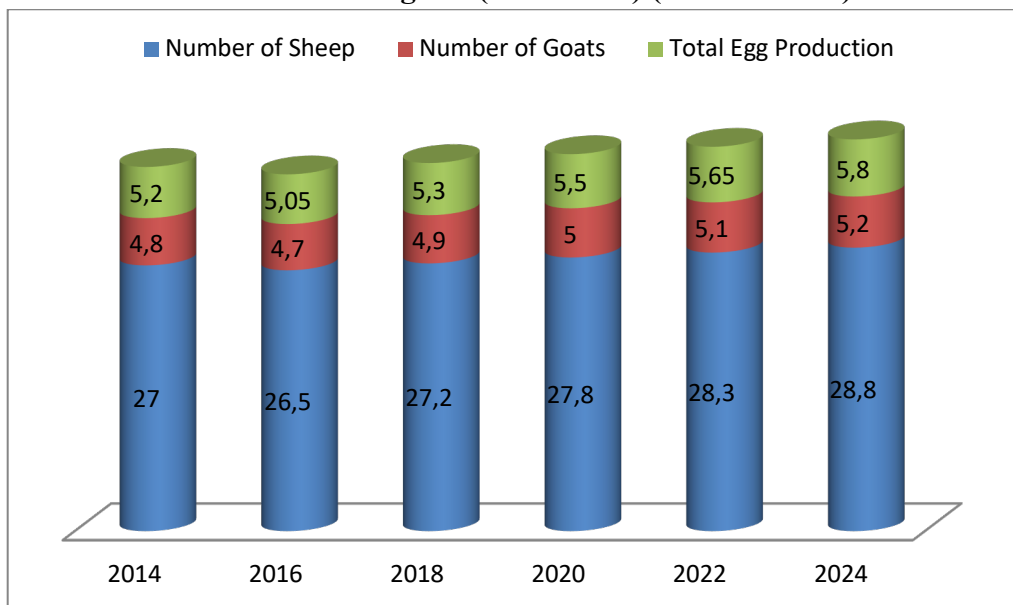
Source: Foreign Agricultural Service, 2022, p15.

Figure (02) illustrates the evolution of the total value of livestock production in Algeria between 2014 and 2022. The sector recorded a significant decline from 8,959 billion DZD in 2014 to 6,676 billion DZD in 2016, mainly due to climatic and economic shocks. This was followed by a gradual recovery, with the value reaching 7,234 billion DZD in 2018 and peaking at 9.5 billion DZD in 2022. This upward trend reflects the sector's revitalization, driven by government support programs and improved productivity. (FAO,2024)

3.2 Biological Dimension

- ✓ The decline in milk production, despite the introduction of improved breeds, highlights persistent challenges related to nutrition, animal health, and infrastructure weaknesses.
- ✓ The increase in sheep milk production demonstrates successful biological adaptation and the emergence of a new value chain contributing to both dietary diversity and economic resilience.

Fig.3. The Biological Dimension of Livestock Development and Egg Production in Algeria (2014–2024) (Million Units)



Source: National Office of Statistics, 2019, p.45

Figure (03) illustrates the biological evolution of sheep and goat herds, as well as egg production in Algeria between 2014 and 2024. The data reveal a relative stability in herd numbers, with minor fluctuations caused by climatic factors and feed shortages, followed by gradual growth supported by genetic improvement programs and pasture expansion.

At the same time, egg production displayed a clear upward trend, reflecting improvements in production practices and a growing reliance on intensive farming systems. This progress highlights the increasing contribution of the livestock sector to achieving national food security. (ONS, 2019)

4. CONCLUSION

Livestock in Algeria constitutes a cornerstone for achieving food security and promoting economic development. It plays a pivotal role in providing animal protein, creating job opportunities, and supporting related agro-industries. Despite the sector's considerable potential, it faces biological, environmental, and economic challenges that require innovative solutions and integrated policies.

Statistical indicators reveal fluctuations in production levels over the last decade: while fresh milk output declined before regaining part of its growth, meat and dairy production showed relative resilience under difficult conditions. The sustainability of this sector depends on a combination of technological development, sound environmental management, and investments in biological research to enhance productivity and improve the quality of animal products.

4.1 Research Findings

- ✓ Local breeds demonstrate high biological importance due to their resistance to diseases and adaptability to harsh climatic conditions, yet their productivity remains low compared to imported breeds.
- ✓ Fresh milk production declined by nearly 19% between 2014 and 2020, before showing slight improvement in recent years.
- ✓ Red meat production has remained relatively stable, with slow growth reflecting gradual improvements in breeding and feeding techniques.
- ✓ Sheep milk production recorded a notable increase until 2020, contributing to food security, particularly in rural areas.
- ✓ Heavy reliance on imported feed, due to the degradation of natural pastures, has increased costs and negatively affected the agricultural trade balance.

4.2 Recommendations and Proposals

- ✓ Develop genetic selection programs for local breeds to enhance productivity while preserving their environmental adaptability.
- ✓ Improve rangeland management through the adoption of sustainable grazing systems and anti-desertification measures to reduce dependence on imported feed.
- ✓ Strengthen biological research in areas such as animal nutrition, genetic improvement, and disease control.

- ✓ Expand veterinary vaccination programs and promote the use of electronic identification systems for herd health monitoring.
- ✓ Encourage private investment in livestock-related processing industries such as dairy, leather, and wool.
- ✓ Integrate modern technologies such as precision agriculture and

artificial intelligence into livestock farm management.

5. Bibliography List:

1.Books

- Ensuring that biological diversity is preserved while meeting the growing demands of food security and economic (Brown, 2006, p. 35).

2.Journal article :

- Dairy Global. (2019, October 17). Algeria's future dairy industry looks bright. Retrieved from Food and Agriculture Organization of the United Nations (FAO). (2024). FAOSTAT: Gross value of livestock production – Algeria.
- Foreign Agricultural Service. (2022). Dairy and products annual: Algeria. United States Department of Agriculture.

3.Internet websites:

- Food and Agriculture Organization of the United Nations (FAO). (2005). Algeria livestock sector policies. Retrieved from <https://www.fao.org/ag/againfo/programmes/en/lead/toolbox/Mixed1/LSP-Algeria.html>
- Husfarm. (n.d.). Agricultural statistics: Algeria. Retrieved from <https://husfarm.com/statistic>
- Foreign Agricultural Service. (2022). Dairy and products annual: Algeria. United States Department of Agriculture.

4.Reports :

- Office National des Statistiques (Algérie). (2019). Rétrospective de l'agriculture en Algérie: productions végétales et animales (2010–2019). Rapport statistique, ONS.

World Bank. (2023). Algeria economic update, spring 2023. World Bank